Tutorial 5 – Derivatives of composite fns. & Inv. Trigon. Fns.
Name: ____________________ **Div. and Roll No: ________________**
Date: __________________

Find $\frac{dy}{dx}$ for the following:

(1) $y=(2x+1)^3$ Ans. $6(2x+1)^2$

(2) $y=\sin^6 x$ Ans. $6\sin^5 x.\cos x$

(3) $y=\sin\left(x^2+5\right)$ Ans. $2x\cos\left(x^2+5\right)$

(4) $y=\sin\left(\cos x^2\right)$ Ans. $-2x\sin x^2.\cos\left(\cos x^2\right)$

(5) $y=\cos x^3.\sin^2 x^5$ Ans. $10x^4\sin x^5\cos x^5\cos x^3-3x^2\sin x^3\sin^2 x^5$

(6) $y=\frac{\cos x}{x}-x.\sin x$ Ans. $\frac{-x\sin x-\cos x}{x^2}-x\cos x-\sin x$

(7) $y=\cos\left(\sqrt{x}\right)$ Ans. $\frac{\sin\left(\sqrt{x}\right)}{2\sqrt{x}}$

(8) $\mathrm{y}=\sec^{-1}\left(\frac{1+x^2}{1-x^2}\right)$ Ans. $\frac{2}{1+x^2}$

(9) $y=\sin^{-1}\left(\frac{1-\cos 2x}{2\sin x}\right)$ Ans.1

(10) $y=\tan(\cot^{-1}\frac{1}{x})$ Ans.1

(11) $y=\sin^{-1}\left(2x\sqrt{1-x^2}\right)$ Ans. $\frac{2}{\sqrt{1-x^2}}$

(12) $y=cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ Ans. $\frac{2}{1+x^2}$

Tutorial 6 – Derivatives of Exponential and Logarithmic Fns.
Name: ____________________ **Div. and Roll No: ________________**
Date: __________________

Find $\frac{dy}{dx}$ for the following:

(1) $y = e^{x^2} + e^{x^3}$ Ans. $2xe^{x^2} + 3x^2e^{x^3}$

(2) $y = e^{\cos x}$ Ans. $-\sin x.e^{\cos x}$

(3) $y = x.7^x$ Ans. $x.7^x \log 7 + 7^x$

(4) $y = e^{\cos^{-1} x}$ Ans. $-\frac{e^{\cos^{-1} x}}{\sqrt{1-x^2}}$

(5) $y = e^{2x^4}$ Ans. $8x^3e^{2x^4}$

(6) $y = \sqrt{e^{\sqrt{x}}}$ Ans. $\frac{e^{\sqrt{x}}}{4\sqrt{xe^{\sqrt{x}}}}$

(7) $y = 5^x - e^x + x^5$ Ans. $5^x \log 5 - e^x + 5x^4$

(8) $y = (\log x)^{\cos x}$ Ans. $(\log x)^{\cos x}\left[\frac{\cos x}{x \log x} - \sin x.\log(\log x)\right]$

(9) $y = x^x - 2^{\sin x}$ Ans. $x^x(1+\log x) - 2^{\sin x}\cos x \log 2$

(10) $y = (x+3)^2.(x+4)^3.(x+5)^4$ Ans. $(x+3).(x+4)^2.(x+5)^3\left[9x^2 + 70x + 133\right]$

(11) $y = \sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}$

Ans. $\frac{1}{2}\sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}\left[\frac{1}{(x-3)} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5}\right]$

(12) If $f(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)$, the find $f'(1)$. Ans. 120

Tutorial 7 – Derivatives of Implicit and Parametric Fns.
Name: ____________________ **Div. and Roll No: ________________**
Date: __________________

(1) Find $\frac{dy}{dx}$, if $x - y = \pi$ Ans. 1

(2) Find $\frac{dy}{dx}$, if $x^{2/3} + y^{2/3} = a^{2/3}$ Ans. $-\frac{y^{1/3}}{x^{1/3}}$

(3) Find $\frac{dy}{dx}$, if $y + \sin y = \cos x$ Ans. $\frac{\sin x}{1+\cos y}$

(4)Find $\frac{dy}{dx}$, if $y^4 + x^5 - 7x^2 - 5x^{-1} = 0$ Ans. $\frac{-5x^4 + 14x - 5x^{-2}}{4y^3}$

(5) Find $\frac{dy}{dx}$, if $\sin y + x^2 y^3 - \cos x = 2y$ Ans. $\frac{2xy^3 + \sin x}{2 - \cos y - 3x^2 y^2}$

(6) Find $\frac{dy}{dx}$, if $x = a(\theta - \sin\theta); y = a(1+\cos\theta)$ Ans. $-\cot\frac{\theta}{2}$

(7) Find $\frac{dy}{dx}$, if $x = 5t; y = \frac{5}{t}$ Ans. $-\frac{1}{t^2}$

(8) Differentiate $x.\cos x$ w.r.t. $\tan x$. Ans. $\frac{\cos x - x.\sin x}{\sec^2 x}$

(9) Find $\frac{dy}{dx}$, if $x = 2at^2, y = at^4$ Ans. t^2

(10) Find $\frac{dy}{dx}$, if $x = \sin t, y = \cos 2t$ Ans. $-4\sin t$

Tutorial 8 – Successive Differentiation & Applications of Derivatives.
Name: ____________________ **Div. and Roll No: ________________**
Date: __________________

(Second Order Derivatives, Increasing and decreasing functions, Maxima and Minima, Concavity and Points of Inflection)

(1)) Find $\frac{d^2y}{dx^2}$, if $y = \tan^{-1} x$ Ans. $\frac{-2x}{\left(1+x^2\right)^2}$

(2) Find $\frac{d^2y}{dx^2}$, if $y = \sqrt{x}$ Ans. $-\frac{1}{4x^{3/2}}$

(3) If $y = e^{2\sin^{-1} x}$, show that $(x^2 - 1)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 4y = 0$

(4) Find $\frac{d^2y}{dx^2}$, if $y = x^3 \log x$ Ans. $x(5 + 6\log x)$

(5) A stone is dropped into a quiet lake and waves move in the form of circles at a speed of 4 cm/sec. At the instant, when the radius of the circular wave is 10 cm, how fast is the enclosed area increasing?

(6) Show that the function $f(x) = x^3 - 6x^2 + 12x - 18$ is increasing for all values of x .

(7) State the intervals on which f(x) is increasing or decreasing, where
$f(x) = 2x^3 - 3x^2 - 36x + 7$

Ans. $f(x)$ is increasing for $x \in (-\infty, -2)$ and $x \in (3, \infty)$ & decreasing for $x \in (-2, 3)$

(8) Show that the function $f(x) = x^3 - 3x^2 + 3x - 10$ is increasing on R .

(9) Determine the maximum and minimum values of the function $f(x) = x^5 - 5x^4 + 5x^3 - 1$

Ans. Max value = 0 at $x = 1$ and min value = -28 at $x = 3$

(10) Determine the maximum and minimum values of the function $f(x) = 2x^3 - 9x^2 + 12x - 3$

Ans. Max value = 2 at $x = 1$ and min value = 1 at $x = 2$